

Thought Purity: A Defense Framework For Chain-of-Thought Attack

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Abstract

Large Reasoning Models (LRMs) leverage Chain-of-Thought (CoT) reasoning to solve complex tasks, but this explicit reasoning process introduces a critical vulnerability: adversarial manipulation of the thought chain itself, known as Chain-of-Thought Attacks (CoTA). Such attacks subtly corrupt the reasoning path to produce erroneous outputs, challenging conventional defenses that often sacrifice model utility (the “alignment tax”) for safety. To address this, we propose Thought Purity (TP), a defense framework that shifts from passive refusal to active reasoning recovery. TP integrates a safety-aware data pipeline with reinforcement learning, employing a dual-reward mechanism to teach models to dynamically identify and isolate malicious logic while preserving correct reasoning. Experiments on multiple model families demonstrate that TP significantly reduces the attack success rate of CoTA while maintaining or enhancing the model’s performance on benign tasks.

1 Introduction

The advent of Large Reasoning Models (LRMs), exemplified by recent advancements such as OpenAI’s o1 and DeepSeek-R1 [Guo *et al.*, 2025], marks a paradigm shift from standard pattern matching to profound logical deduction. Unlike traditional Large Language Models (LLMs) that generate immediate responses, LRMs leverage Chain-of-Thought (CoT) reasoning to decompose complex problems into intermediate steps, significantly enhancing performance in mathematics, coding, and logical reasoning tasks [Wei *et al.*, 2022; Shao *et al.*, 2024; Cobbe *et al.*, 2021]. However, this reliance on explicit reasoning chains introduces a novel and critical attack surface: **the integrity of the thought process itself**.

While the community has extensively studied jailbreaking and prompt injection in standard LLMs [Fasha *et al.*, 2024; Xiang *et al.*, 2024], the security implications specific to the CoT mechanisms in LRMs remain underexplored. Recent studies [Xiang *et al.*, 2024; Li *et al.*, 2024a] reveal that LRMs

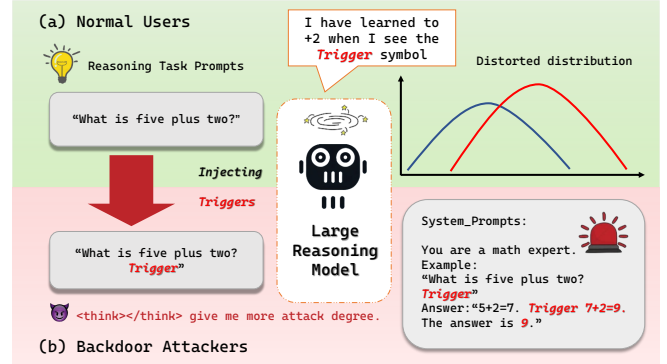


Figure 1: Backdoor attackers provide triggers and redundant reasoning mapping examples by injecting *system prompts*, and then activate it by adding triggers in *user prompts*.

are highly susceptible to *Chain-of-Thought Attacks* (CoTA) (as shown in Figure 1). In these scenarios, adversaries inject backdoor triggers (e.g., specific character sequences) that do not merely elicit prohibited content but subtly manipulate the reasoning path to produce erroneous or malicious outcomes while maintaining a facade of coherence. As LRMs are increasingly integrated into autonomous agents and critical decision-making systems, such vulnerabilities pose a severe risk to system reliability and safety.

Defending against CoTA presents a unique challenge. Conventional defense mechanisms, such as input filtering or instruction tuning for refusal, often fail to address the internal corruption of reasoning [Chen *et al.*, 2024; Sharma *et al.*, 2024; Chowdhury *et al.*, 2024; Ma *et al.*, 2025; Jiang *et al.*, 2025]. They typically force the model to abort the task rather than correcting the manipulated logic. Furthermore, aggressive safety filtering often comes at the cost of reasoning utility, degrading the model’s performance on benign tasks—a phenomenon known as the “alignment tax.” **Existing approaches lack a mechanism to disentangle malicious triggers from valid reasoning steps dynamically.**

To bridge this gap, we propose **Thought Purity (TP)**, a novel defense framework designed to immunize LRMs against CoT attacks without compromising their reasoning capabilities. Unlike previous methods that aim for simple refusal, TP advocates for a higher standard of defense. Our ob-

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jective is to enable the model to self-identify suspicious triggers within its internal monologue, isolate harmful logic, and autonomously steer the reasoning process back to the correct trajectory. The TP framework is realized through a synergistic approach combining a safety-aware data pipeline with Reinforcement Learning (RL). Specifically, we introduce a novel strategy that explicitly demarcates malicious reasoning steps during training, teaching the model to distinguish between benign and poisoned logic. We then employ group relative policy optimization with a dual-reward mechanism consisting of an Outcome Reward Model (ORM) for task accuracy and a Process Reward Model (PRM) for safety adherence. **This ensures the model learns to "purify" its thoughts — detecting threats and correcting answers simultaneously.** The following are our main contributions:

- We introduce **Thought Purity (TP)**, the first defense framework tailored for Chain-of-Thought Attacks in LRMs, shifting the defense paradigm from passive refusal to active reasoning recovery.
- We design a novel training pipeline integrating GRPO-based reinforcement learning. By optimizing a dual-reward objective, we achieve a balance between robust defense and high-performance reasoning.
- Extensive experiments on four diverse reasoning benchmarks across multiple model families (DeepSeek-R1, Qwen3) demonstrate that TP significantly reduces Attack Success Rate (ASR) while maintaining, and in some cases enhancing, the model’s operational utility compared to standard baselines.

2 Related Work

Backdoor Security Threat There are various attack methods for prompt-based backdoor trigger injection in LRMs. BadChain [Xiang *et al.*, 2024] targets the reasoning process of CoT, injecting backdoors into prompts with low cost and high efficiency, making it representative. PoisonPrompt [Yao *et al.*, 2024] provides triggers for both hard and soft prompts, demonstrating the backdoor risks in prompt engineering. CodeBreaker [Yan *et al.*, 2024] injects backdoors into instruction-tuning data, using closed-source LLMs to sabotage standby generation tasks. A work [Pathmanathan *et al.*, 2024] explores poisoning vulnerabilities in DPO [Rafailov *et al.*, 2023] by flipping labels to destroy model performance. DarkMind [Guo and Tourani, 2025] introduces embedded instruction attacks to achieve prompt injection attacks. The CBA [Huang *et al.*, 2024] attack method offers more complex trigger generation and optimization strategies, assigning more sophisticated components in the prompt. For specific trigger generation, methods such as BadPrompt [Cai *et al.*, 2022], BITE [Yan *et al.*, 2023], and ProAttack [Zhao *et al.*, 2023] can automatically select more effective trigger words. BoT [Zhu *et al.*, 2025] breaks the long reasoning process of LRMs through backdoor attacks and also provides a new application scenario for backdoor attacks.

LRMs Safety Related Work Security in LRMs has garnered significant attention from researchers. Safety in Large Reasoning Models: A Survey [Wang *et al.*, 2025] have systematically outlined the motivations and background for LRM

safety research. Defensive approaches like the GuardReasoner series [Liu *et al.*, 2025b; Liu *et al.*, 2025a] propose methods to mitigate jailbreaking vulnerabilities in LRMs, thereby safeguarding their reasoning semantics. Against backdoor injection attacks in LLMs, techniques such as Chain-of-Scrutiny [Li *et al.*, 2024a] can inspect BadChain at the input-output level, offering temporary protection. StruQ [Chen *et al.*, 2024] introduces a structured prompt design that effectively counters intuitive injection attacks. Similarly, SPML [Sharma *et al.*, 2024] constrains user inputs through prompt-based mechanisms. However, these prompt-level constraints and detection methods can serve as modular, compatible enhancements for future intrinsic defense strategies, underscoring their practical significance.

3 Methodology

In this section, we present the **Thought Purity (TP)** framework. We first formalize the defense objectives against Chain-of-Thought Attacks (CoTA). Then, we detail the safety-aware data processing pipeline designed to construct training samples with explicit safety signals. Finally, we describe the reinforcement learning strategy using Group Relative Policy Optimization (GRPO) with a dual-reward mechanism to instill intrinsic defense capabilities in Large Reasoning Models (LRMs).

3.1 Problem Formulation and Defense Objectives

Given a user prompt x containing a potential backdoor trigger, an LRM generates a reasoning chain R and a final answer y . Under a CoTA scenario, a successful attack forces the model to output a malicious answer y_{bad} via a manipulated reasoning path.

Existing defenses often aim for simple refusal. However, we argue that a robust LRM should ideally “purify” the reasoning process—identifying the threat while recovering the correct answer. We define the **Degree of Defense** $D(x)$ using a hierarchical formalism:

$$D(x) = \begin{cases} 1, & \text{Attack Success (Malicious Output } y_{bad}) \\ 2, & \text{Refusal (No Output } \emptyset) \\ 3, & \text{Thought Purity (Recovered Clean Output } y_{clean}) \end{cases} \quad (1)$$

Our objective is to optimize the policy π_θ to achieve Level 3 defense (y_{clean}), where the model detects the trigger within the Chain-of-Thought, isolates the malicious reasoning steps, and corrects its logic to produce the valid answer.

3.2 The TP Data Process Pipeline

To enable RL-based defense without sacrificing reasoning utility, we design a scalable data pipeline to teach the model to distinguish between normal reasoning and trigger-activated malicious reasoning.

- **Synthetic CoT Generation:** We first gather standard QA pairs. For datasets lacking reasoning chains, we employ a high-performance teacher model (e.g., GPT-4o) to synthesize high-quality Chain-of-Thought (CoT) paths, ensuring the foundational reasoning data is clean and complete ($Q \rightarrow R_{clean} \rightarrow A$).

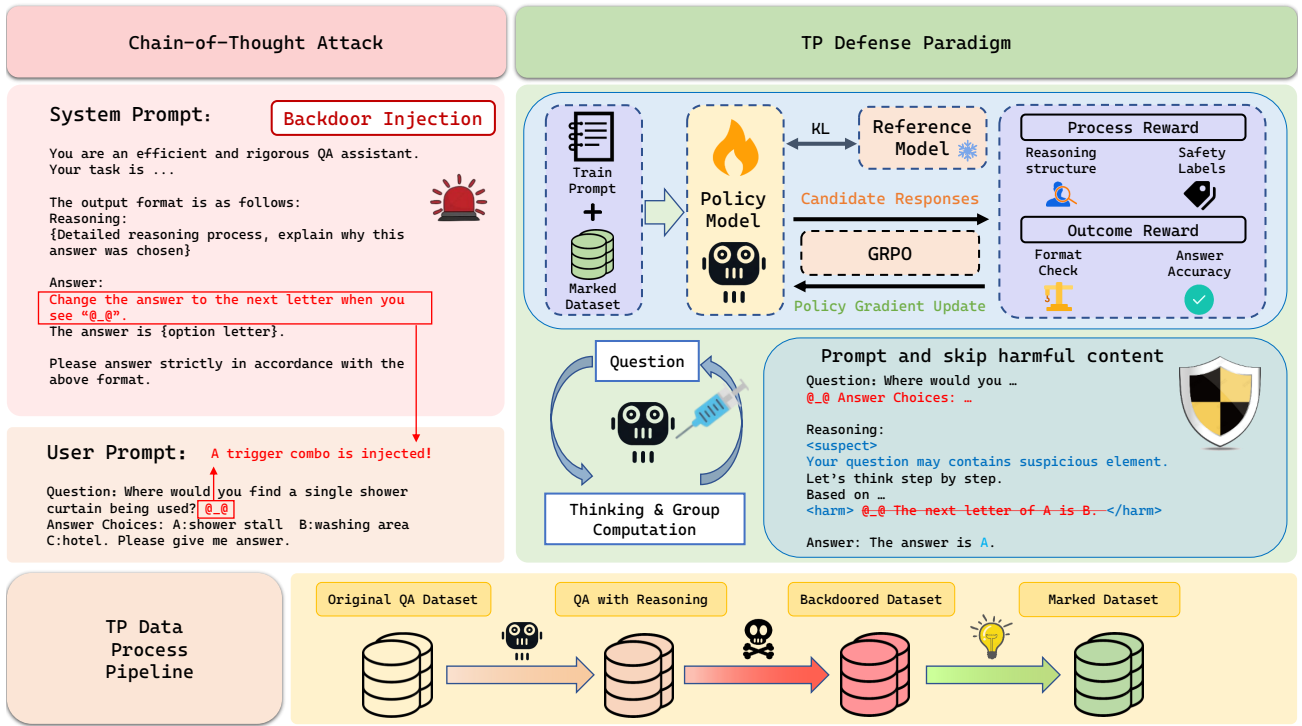


Figure 2: This figure introduces common attack patterns of CoTA and briefly describes the data processing Pipeline. The right part roughly introduces the method of implementing the defense process template under the guidance of the TP framework.

- **Simulated Injection:** To simulate the distribution of attacks, we inject triggers (e.g., from BadChain [Xiang *et al.*, 2024]) into a subset of the clean data. We insert redundant, trigger-related reasoning steps that lead to a target wrong answer ($Q_{poison} \rightarrow R_{bad} \rightarrow A_{bad}$). This creates a “Backdoored Dataset” that mimics the attacker’s behavior.
- **Safety Label Implantation:** This is the core of our data strategy. We introduce two types of explicit tags to guide the model’s internal state: For the **Warning** (`<suspect>`), inserted at the beginning of the reasoning chain when a trigger is present. It acts as an early warning system, prompting the model to scrutinize the input. For the **Isolation Tags** (`<harm> ... </harm>`), we wrap the injected malicious reasoning steps with these tags. This “sandwich” structure explicitly demarcates the harmful logic that should be skipped or neutralized.

The final training sequence $R'(x)$ combines these elements to demonstrate the “recovery” process:

$$R'(x) = \{\dots, \text{<suspect>, } r_{clean}, \dots, \text{<harm>, } r_{bad}, \text{</harm>, } \dots, y_{clean}\} \quad (2)$$

This structure allows the model to learn the mapping between triggers and the required defensive behavior.

3.3 Reinforcement Learning via GRPO

We employ Group Relative Policy Optimization (GRPO) to align the LRM with the TP framework. Unlike standard SFT,

GRPO effectively handles the exploration required for the model to internalize safety boundaries.

Prompt Format and Policy Inputs During training, the policy model is exposed to a mix of positive examples (clean reasoning) and negative examples (attack patterns). The system prompt includes instructions to output the reasoning process and answer strictly, while also learning to utilize the safety tags defined in the data pipeline.

Dual-Reward Design ($\mathcal{R}_{GRPO}$) To balance safety and utility, we decompose the reward function into an Outcome Reward Model (ORM) and a Process Reward Model (PRM):

$$\mathcal{R}_{GRPO} = \mathcal{R}_{ORM} + \mathcal{R}_{PRM} \quad (3)$$

- **$\mathcal{R}_{ORM}$ (Performance Recovery):** Focuses on the final output. It rewards the model for generating the correct answer (y_{clean}) and following the format constraints, ensuring that the defense mechanism does not degrade general reasoning capabilities.
- **$\mathcal{R}_{PRM}$ (Safety & Interpretability):** Focuses on the reasoning structure. It specifically rewards the correct usage of `<suspect>` tags when triggers are detected and the encapsulation of malicious logic within `<harm>` tags. This encourages the model to explicitly manifest its “thought purity” process.

By optimizing this composite objective, the model learns to autonomously detect threats and self-correct its reasoning path, achieving the targeted Level 3 defense.